

SHREYA SINHA

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Summary

Backend engineer moving into product, with a habit of framing technical decisions as user and business trade-offs. Brings data fluency and close rapport with engineering teams to discovery, prioritisation and outcome-driven delivery.

Education

Master of Computer Applications (MCA) <i>Birla Institute of Technology, Mesra</i>	CGPA: 8.23 2024 – 2026
Bachelor of Computer Applications (BCA) <i>Patna Women's College — Platinum Jubilee Scholarship, all six semesters</i>	CGPA: 9.01 2021 – 2024

Technical and Product-Analytical Experience

Network People Services Technologies (NPST) Jul 2026 – Present

- Onboarded onto a **live UPI/NPCI payments platform** serving real merchant and consumer transactions, and **shipped production features within the first month** — working directly on money-movement infrastructure from day one.
- Surfaced and **resolved transaction data discrepancies** across multiple services and databases before release, **protecting transaction integrity** for end users and preventing defects from reaching production payment flows.
- Partnered with Product, QA, and Business stakeholders** in short Agile cycles, presenting data findings and blockers in sprint reviews to **keep releases on schedule** — the same cross-functional rhythm an APM operates in.

Projects

Fraud Detection System | *Python, FastAPI, Data Pipelines, Anomaly Detection* Jan 2026 – Feb 2026

- Framed and resolved a precision/recall trade-off** for a fraud-detection model on a **2% positive class (~100K transactions)** — the real decision fraud and risk teams needed, not just “best accuracy.”
- Compared three anomaly-detection approaches** (Isolation Forest, One-Class SVM, Autoencoders) on Precision, Recall, F1, and AUC-ROC, then **shipped the selected model as a documented, production-ready REST API** with request validation and structured logging.

Smart AI Traffic Management System | *YOLOv8, PyTorch, Flask, LLaMA 3* Feb 2026 – Mar 2026

- Trained a **Deep Q-Network** to control 3 traffic signal phases from a real-time computer-vision pipeline (100+ detections/frame, ~90% accuracy), **cutting average wait time 30% and raising throughput 25%** vs. a fixed-timer baseline.
- Recognized that **operators would not trust decisions they could not interrogate**, and integrated a locally hosted LLaMA 3 layer that translated every signal decision into **plain-language explanations** — the difference between a working model and an adopted one.

Research Publications

- Journal:** S. Sinha, L. Hota, K. K. Senapati, and A. Kumar, “SmartTraffic AI: A Real-Time Adaptive Traffic Signal Control System,” *IEEE Transactions on Networking*, 2026. [**Communicated**]
- Conference:** S. Sinha, L. Hota, and K. K. Senapati, “SmartTraffic AI for a Real-Time Adaptive Traffic Signal Control System,” *2026 IEEE Future Networks World Forum (FNWF)*, 2026. [**Communicated**]
- Conference:** S. Sinha, L. Hota, K. K. Senapati, and P. Paul, “A Real-Time AI-Driven Traffic Management System using Deep Vision and Context-Aware AI,” *4th International Conference on Innovation in Engineering & Management (ICIEM'26)*, BIT Noida, 2026. [**Accepted**]

Leadership & Volunteering

Volunteer Teacher — School Mentorship Program, Patna Women's College

- Volunteered to close learning gaps at a government all-girls school, **owning instruction for 120+ students** across Classes 7–11 with widely varying learning levels.
- Adapted lesson plans to each grade's level and pace and iterated on delivery using classroom feedback, **improving the student pass rate by 10%**.

Skills

Product & Analysis: Metric Definition & Trade-off Analysis, Success Criteria, Technical Documentation, Communicating Technical Work to Non-Technical Audiences, Data Analysis (SQL)

Languages & Data: Python, SQL, JavaScript, Java, C++; PostgreSQL, MySQL, MongoDB

Backend & Systems: FastAPI, Flask, REST APIs, Spring Boot, React, Linux, Git, Docker, Postman

Data & ML: Data Pipelines, Feature Engineering, PyTorch, Scikit-learn, Anomaly Detection, Computer Vision, Reinforcement Learning, NLP